

Ashfak Md Shibli

Software Engineer, AI Systems & Full-Stack Infrastructure
Open to onsite/hybrid relocation | 7+ YOY

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EXPERIENCE

- Athlete Den** New York, USA
Software Engineer May 2024 – Current
 - Own backend and AI systems for an early-stage sports analytics platform, including Python/GCP data pipelines, Label Studio training workflows, Firebase real-time sync, CI/CD, crash reporting, and production monitoring.
 - Built and optimized computer vision inference workflows for real-time sports activity detection and performance analytics, improving model performance by 85% through dataset, model, and deployment iteration.
 - Shipped product features across backend, web, mobile, and AI workflows in a high-ownership startup environment, working directly with leadership on product direction, commercialization, and patent-pending technology.
 - Developed Next.js/JavaScript web experiences and cloud-backed product workflows with Firebase analytics, offline caching, background sync, and auditable data traces for user and model behavior.
- New Jersey Institute of Technology + Tennessee Tech University** New Jersey + Tennessee, USA
Graduate Research and Teaching Assistant Sep 2022 – May 2024
 - Built automated PHI/PII leakage detection systems for Android health apps using Python and STRIDE threat modeling; found 61% of evaluated medical/health apps had at least one leak, including 38% with PHI leakage.
 - Developed SMS/LLM threat-analysis pipelines using BERT embeddings, BERTopic clustering, VirusTotal/IPQS/Gemini integrations, and D3 dashboards; achieved approximately 92% detection accuracy.
 - Created security research tooling for adversarial/phishing analysis, dataset processing, threat-intelligence enrichment, explainability, and visualization across Python, SQL, and web-based dashboards.
- Samsung Research** South Korea & Bangladesh
Lead Software Engineer Jul 2018 – Aug 2022
 - Led development of enterprise-scale Samsung Family Hub software with secure layered APIs, cloud synchronization, local data management, and robust JSON/XML processing.
 - Engineered multi-threaded local data systems for offline interaction, data integrity, and fault tolerance, reducing concurrency-related crashes by 99% across production product workflows.
 - Architected multi-platform Samsung Galaxy Buds applications across iOS, Android, Watch, and Windows with BLE byte-stream handling, authenticated REST APIs, and firmware update systems.
 - Built automated testing and observability workflows using WinAppDriver, CI practices, vulnerability/I/O anomaly logging, crash analytics, and performance diagnostics for proactive issue detection.
 - Prepared invention disclosures, evaluated security and novelty implications, created 120+ technical knowledge-base articles, and coordinated engineering work across global product teams.

TECHNICAL SKILLS

- Languages & Systems:** Python, JavaScript/TypeScript, Swift, C#, SQL, Shell Scripting, REST APIs, JSON/XML
- AI/ML & Agents:** LLMs, RAG, agentic coding workflows, Codex/ChatGPT, Claude/Claude Code, Cursor/Copilot-style tools, PyTorch, scikit-learn, BERT, CNNs, model evaluation
- Backend, Data & Cloud:** Google Cloud Platform, Firebase/NoSQL, PostgreSQL, MySQL, BigQuery, pandas, NumPy, data pipelines, event/log analysis
- Frontend & Product:** Next.js, React, JavaScript, HTML/CSS, D3.js, analytics dashboards, mobile/web product workflows
- DevOps, Testing & Observability:** Docker, GitHub Actions CI/CD, Git, pytest, XCTest, WinAppDriver, TestFlight, Firebase Analytics, crash reporting
- Security & Healthcare:** Threat modeling, PHI/PII leakage detection, STRIDE, adversarial/LLM security, smishing detection, HIPAA-aware data protection

EDUCATION

- New Jersey Institute of Technology + Tennessee Technological University**
M.S. Computer Science (GPA: 4.00/4.00) 2022 – 2024

PATENTS, PUBLICATIONS & RECOGNITION

- Patent:** Application-pending “System, Apparatus, and Method for Analyzing Sports Performance Using Multi-Media Content.”
- Selected Publications:** “SmishViz: Graph-based Visualization System for Smishing Threats,” ACM CODASPY 2025 (20.9% acceptance) | “AbuseGPT: Abuse of Generative AI ChatBots to Create Smishing Campaigns,” IEEE ISDFS 2024.
- Recognition:** Eminence Award 2025, Athlete Den Employee of the Year, Samsung R&D ICON of the Month, Samsung Outstanding Collaborator.